

MODULE-4 SOIL POLLUTION & SOLID WASTE MANAGEMENT

The Earth is divided into 4 “Subsystems” or “Spheres” namely:

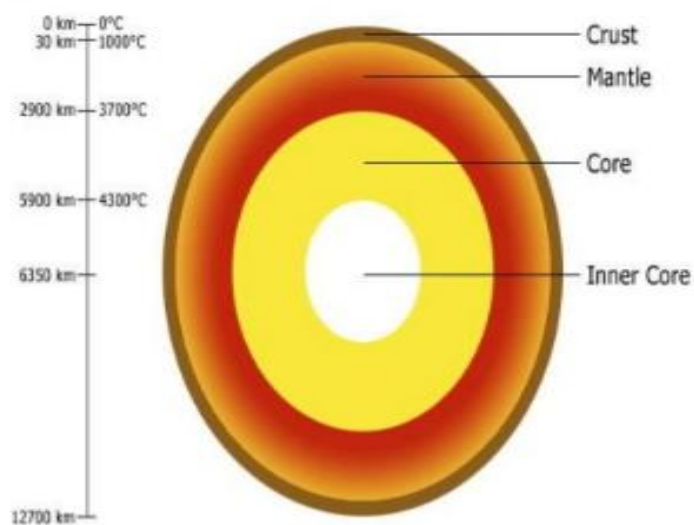
ATMOS(Gaseous)-SPHERE, HYDRO(Water)- SPHERE , LITHOS(Rocky)-SPHERE and BIOS(Life)-SPHERE

They are interdependent and function as a whole.

Lithosphere

The Lithosphere is the top, rigid layer of the earth consisting of crust and upper mantle. It is primarily the rocky and stony part covering the whole earth surface ; on which the interaction of the other Geospheres happen.

STRUCTURE OF LITHOSPHERE



1. Oceanic Lithosphere : It consists mainly of mafic (rich in magnesium and iron) crust and ultramafic(over 90% mafic) mantle and is denser than continental lithosphere.
2. Continental Lithosphere : It is also called the Continental crust. It is the layer of igneous, sedimentary rock that forms the continents and the continental shelves. This layer consists mostly of granitic rock.
3. CRUST: It is the outermost layer of the earth with average density of 2.8- 3.0 g/cm³ and average thickness of 30 km. It consists of the continental crust and the oceanic crust. Life exists in this layer.
4. MANTLE: It is the second layer of the earth and extends from 30km- 2900km with an average density of 2.7 g/cm³ . It contains 83% of the total volume and 68% of the total mass of the earth. It is made up of silicate rich iron and magnesium and is divided from the crust by a discontinuity and MOHOROVICIC DISCONTINUITY.
5. CORE: It is the deepest layer of earth. It extends from a distance of 2900km – 6371 km with an average density of 4.3- 11 g/cm³ . Mantle and core boundary is called WEICHERT- GUTENBERG Discontinuity. Core volume is 16% of earth's total volume and core mass is 32% of earth's total mass. It is further divided into 2 sub-zones by the discontinuity called as LEHMANN- BULLEN Discontinuity.

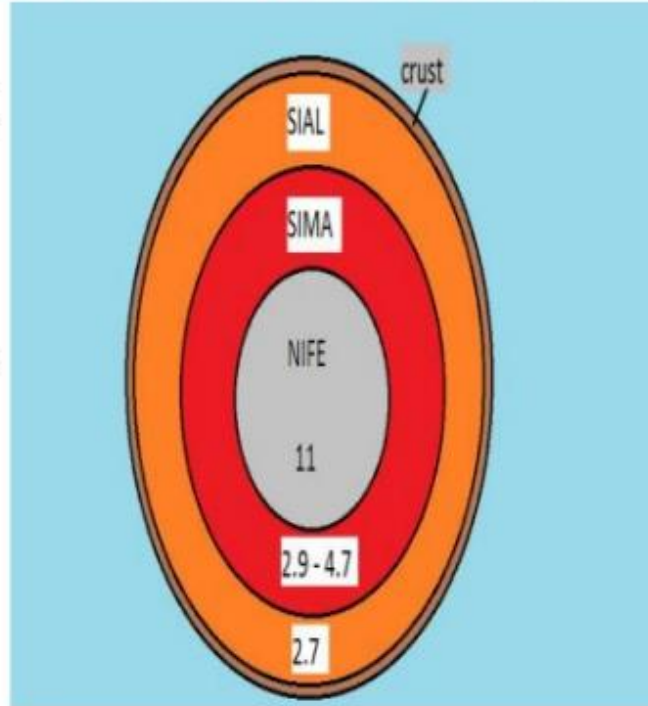
OUTER CORE

INNER CORE

CHEMICAL COMPOSITION OF LITHOSPHERE

According to E. Seuss the earth's layers can be broadly divided into 3 types on the basis of chemical composition:

- **SIAL**- located below crust, average density is 2.7 g/cm^3 , thickness ranges between 50-300km; dominated by acid materials.
- **SIMA**- located below sialic layer, average density is $2.9- 4.7 \text{ g/cm}^3$, thickness ranges between 1000- 2000km; dominated by basic matter
- **NIFE**- made up of heavy metals, avg. density is 11 g/cm^3 . Diameter is 6880 km. Magnetic property.



Soil and its Composition

Definition: Soil is a natural body comprised of solids (minerals and organic matter), liquid, and gases that occurs on the land surface, occupies space, and is characterized by one or both of the following: horizons, or layers, that are distinguishable from the initial material as a result of additions, losses, transfers, and transformations of energy and matter or the ability to support rooted plants in a natural environment.

COMPOSITION OF SOIL

Soil is a complex body composed of five major components

- a. mineral matter obtained by the distintergration and decomposition of rocks;
- b. organic matter, obtained by the decay of plant residues, animal remains and microbial tissues;
- c. water, obtained from the atmosphere and the reactions in soil (chemical, physical and microbial);
- d. air or gases, from atmosphere, reactions of roots, microbes and chemicals in the soil

- e. organisms, both big (worms, insects) and small (microbes)
- f. Table I. Classification of soil particles according to two systems (U.S.D.A and International)

Soil separates	U.S. Dept. of Agric. system Diameter (mm)	International system Diameter (mm)	Number of particles per g
Very coarse sand	2.00 - 1.00	)	90
Coarse sand	1.00 - 0.50	) 2.00 - 0.20	720
Medium sand	0.50 - 0.25	)	5,700
Fine sand	0.25 - 0.10	)	46,000
Very fine sand	0.10 - 0.05	) 0.20 - 0.02	722,000
Silt	0.05 - 0.002	)	
Silt		0.02 - 0.002	5,776,000
Clay	below 0.002	below 0.002	90,260,853,000

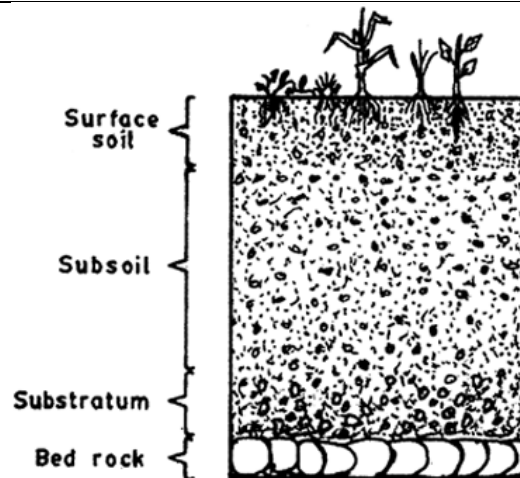


Fig. Soil profile

Soil Physical Properties

All soils contain mineral particles, organic matter, water and air. The combinations of these determine the soil's properties – its texture, structure, porosity, chemistry and colour.

Physical Properties- 1.Horizonation, 2. Soil color, 3. Soil texture, 4. Soil structure, 5.Soil consistence, 6. Bulk density, 7. Pore Space, 8. Soil Permeability, 9. Particle Density

Soil texture

Soil texture refers to the proportion of the soil “separates” that make up the mineral component of soil. These separates are called sand, silt, and clay. These soil separates have the following size ranges:

- Sand = <2 to 0.05 mm
- Silt = 0.05 to 0.002 mm
- Clay = <0.002 mm

Sand and silt are the “inactive” part of the soil matrix, because they do not contribute to a soil’s ability to retain soil water or nutrients. These separates are commonly composed of quartz or some other inactive mineral.

Because of its small size and sheet-like structure, clay has a large amount of surface area per unit mass, and its surface charge attracts ions and water. Because of this, clay is the “active” portion of the soil matrix.

Soil texture can influence whether soils are free draining, whether they hold water and how easy it is for plant roots to grow.

- Sand particles are quite big. The pore spaces between the particles in sandy soils are also quite large. This allows water to drain quickly and air to enter the soil. Sandy soils tend not to get waterlogged in winter but can be subject to drought during summer.
- Silt particles are too small for us to see with our eyes. Silt soils have much smaller pore spaces but a lot more of them.
- Clay particles are smaller than 0.002 mm in diameter. Clay soils are poorly drained and hold on to the water in their pore spaces for much longer. However, they can become very hard if they dry out.

Soil structure

The soil separates can become aggregated together into discrete structural units called “peds”. These peds are organized into a repeating pattern that is referred to as soil structure. Between the peds are cracks called “pores” through which soil air and water are conducted. Soil structure is most commonly described in terms of the shape of the individual peds that occur within a soil horizon. Soil structure is important for plant growth, regulating the movement of air and water, influencing root development and affecting nutrient availability.

Structure is defined in terms of grade, class and type of aggregates.

Grade: Grade of structure is the degree of aggregation and expresses the differential between cohesion within aggregates and adhesion between aggregates. These properties vary with the moisture content of the soil and it should be determined when the moisture content is normal -

not when unusually dry or unusually wet. The four major grades of structure rated from 0 to 3 are listed below.

- 0 Structureless: no observable aggregation or no definite orderly arrangement of natural lines of weakness. Massive if coherent; single grain if noncoherent.
- 1 - Weak: That degree of aggregation characterised by poorly formed indistinct aggregates that are barely observable in place. When disturbed, soil material that has this grade of structure breaks into a mixture of few entire aggregates many broken aggregates and much unaggregated material.
- 2 - Moderate: Well formed distinct aggregates that are moderately durable and evident but not distinct in undisturbed soil. When disturbed, they break down into a mixture of many distinct entire aggregates, some broken aggregates and little unaggregated material.
- 3 - Strong: Durable aggregates that are quite evident in undisturbed soil that adhere weakly to one another. When removed from the profile the soil material consists very largely of entire aggregates and includes few broken ones and little or no nonaggregated material.

Soil colour

In well aerated soils, oxidized or ferric (Fe^{+3}) iron compounds are responsible for the brown, yellow, and red colors you see in the soil. When iron is reduced to the ferrous (Fe^{+2}) form, it becomes mobile, and can be removed from certain areas of the soil. When the iron is removed, a gray color remains, or the reduced iron color persists in shades of green or blue. Upon aeration, reduced iron can be reoxidized and redeposited, sometimes in the same horizon, resulting in a variegated or mottled color pattern. These soil color patterns resulting from saturation, called “redoximorphic features”, can indicate the duration of the anaerobic state, ranging from brown with a few mottles, to complete gray or “gleization” of the soil. Soils that are dominantly gray with brown or yellow mottles immediately below the surface horizon are usually hydric.

Soil color is typically described using some form of color reference chart, such as the **Munsell Color Chart**. Using the Munsell system, color is described in reference to the color’s “hue”, “value”, and “chroma”. Hue describes where in the color spectrum the soil color exists, which for soils includes the colors yellow, red, blue, green, and gray. Value describes the lightness of the color. Chroma indicates the strength of the color. In a Munsell notation, the color is written in the order hue-value-chroma. The color “5YR 4/3” is an example of a Munsell notation, where 5YR is the hue, 4 is the value, and 3 is the chroma.

Soil Consistence

Soil consistency refers to the ease with which an individual ped can be crushed by the fingers. Soil consistence, and its description, depends on soil moisture content. It is the resistance of a soil to deformation or rupture and is determined by the cohesive and adhesive properties of the soil mass. This is a term used to designate the manifestation of the cohesive and adhesive properties of soil at various moisture contents. A knowledge of the consistence of the soil is important in tillage operations, traffic and pond constructions. Consistence gives also an indication of the soil texture.

Consistence is described for three moisture levels:

1. - Wet soil - non sticky, slightly sticky, sticky, very sticky; non plastic, slightly plastic, plastic and very plastic.
2. - Moist soil - loose, very friable, friable, firm, very firm, extremely firm.
3. - Dry soil - loose, soft, slightly hard, hard, very hard, extremely hard.

Bulk Density

Bulk density is the proportion of the weight of a soil relative to its volume. It is expressed as a unit of weight per volume, and is commonly measured in units of grams per cubic centimeters (g/cc).

Bulk density is an indicator of the amount of pore space available within individual soil horizons, as it is inversely proportional to pore space.

Horizonation

Soil “horizons” are discrete layers that make up a soil profile. They are typically parallel with the ground surface. In some soils, they show evidence of the actions of the soil forming processes.

Pore Space:

$$\% \text{ Pore space} = 100 - \frac{\text{bulk density}}{\text{particle density}} \times 100$$

Particle density can be determined using specific gravity bottle technique and bulk density by taking soil core samples of known volume in the field and determining the oven dry weight (Black *et al.*, 1965a). Water and air movements through soil depends on the pore space and the size distribution of the pores (microPores and macropores). Lower the pore space or higher the bulk density of the soil, the higher the suitability of the soil for aquaculture.

Particle density

of soil is the mass per unit volume of soil particles (soil solid phase) - expressed in g/c.c. Most soils have particle density of about 2.6 g/cc. Presence of organic matter decrease the density and iron compounds increase the density.

Soil Permeability:

is the ability of the soil to transmit water and air. An impermeable soil is good for aquaculture as the water loss through seepage or infiltration is low. As the soil layers or horizons vary in their characteristics, the permeability also differs from one layer to another. Pore size, texture, structure and the presence of impervious layers such as clay pan determines the permeability of a soil. Clayey soils with platy structures have very low permeability.

Permeability is measured in terms of permeability rate or coefficient of permeability (cm per hour, cm per day, cm per sec.).

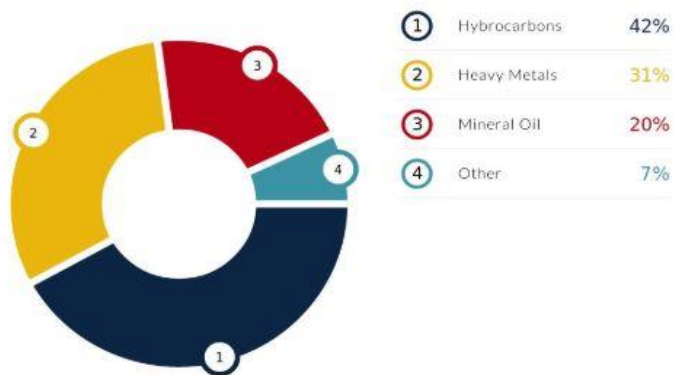
SOIL POLLUTION

Soil pollution is defined as the presence of toxic chemicals (pollutants or contaminants) in soil, in high enough concentrations to pose a risk to human health and/or the ecosystem. In the case of contaminants which occur naturally in soil, even when their levels are not high enough to pose a risk, soil pollution is still said to occur if the levels of the contaminants in soil exceed the levels that should naturally be present.

The root cause of soil pollution is often one of the following:

- Agriculture (excessive/improper use of pesticides)
- Excessive industrial activity
- Poor management or inefficient disposal of waste

An illustration detailing major soil pollutants is provided below.



Main Causes of Soil Pollution

1. Industrial Activity

Industrial activity has been the biggest contributor to the problem in the last century, especially since the amount of mining and manufacturing has increased. Most industries are dependent on extracting minerals from the Earth. Whether it is iron ore or coal, the by-products are contaminated and they are not disposed of in a manner that can be considered safe. As a result, the industrial waste lingers in the soil surface for a long time and makes it unsuitable for use.

2. Agricultural Activities

Chemical utilization has gone up tremendously since technology provided us with modern pesticides and fertilizers. They are full of chemicals that are not produced in nature and cannot be broken down by it. As a result, they seep into the ground after they mix with water and slowly reduce the fertility of the soil. Other chemicals damage the composition of the soil and make it easier to erode by water and air. Plants absorb many of these pesticides and when they decompose, they cause soil pollution since they become a part of the land.

3. Waste Disposal

Finally, a growing cause for concern is how we dispose of our waste. While industrial waste is sure to cause contamination, there is another way in which we are adding to the pollution. Every human produces a certain amount of personal waste products by way of urine and feces. While much of it moves into the sewer the system, there is also a large amount that is dumped directly into landfills in the form of diapers. Even the sewer system ends at the landfill, where the biological waste pollutes the soil and water. This is because our bodies are full of toxins and chemicals which are now seeping into the land and causing pollution of soil.

4. Accidental Oil Spills

Oil leaks can happen during storage and transport of chemicals. This can be seen at most of the fuel stations. The chemicals present in the fuel deteriorates the quality of soil and make them unsuitable for cultivation. These chemicals can enter into the groundwater through the soil and make the water undrinkable.

5. Acid Rain

Acid rain is caused when pollutants present in the air mix up with the rain and fall back on the ground. The polluted water could dissolve away some of the important nutrients found in soil and change the structure of the soil.

Effects of Soil Pollution

1. Effect on Health of Humans

Considering how soil is the reason we are able to sustain ourselves, the contamination of it has major consequences on our health. Crops and plants grown on polluted soil absorb much of the pollution and then pass these on to us. This could explain the sudden surge in small and terminal illnesses. Long term exposure to such soil can affect the genetic make-up of the body, causing congenital illnesses and chronic health problems that cannot be cured easily. In fact, it can sicken the livestock to a considerable extent and cause food poisoning over a long period of time. The soil pollution can even lead to widespread famines if the plants are unable to grow in it.

2. Effect on Growth of Plants

The ecological balance of any system gets affected due to the widespread contamination of the soil. Most plants are unable to adapt when the chemistry of the soil changes so radically in a short period of time. Fungi and bacteria found in the soil that bind it together begin to decline, which creates an additional problem of soil erosion. The fertility slowly diminishes, making land unsuitable for agriculture and any local vegetation to survive. The soil pollution causes large tracts of land to become hazardous to health. Unlike deserts, which are suitable for its native vegetation, such land cannot support most forms of life.

3. Decreased Soil Fertility

The toxic chemicals present in the soil can decrease soil fertility and therefore decrease in the soil yield. The contaminated soil is then used to produce fruits and vegetables which lacks quality nutrients and may contain some poisonous substance to cause serious health problems in people consuming them.

4. Toxic Dust

The emission of toxic and foul gases from landfills pollutes the environment and causes serious effects on the health of some people. The unpleasant smell causes inconvenience to other people.

5. Changes in Soil Structure

The death of many soil organisms (e.g. earthworms) in the soil can lead to alteration in soil structure. Apart from that, it could also force other predators to move to other places in search of food.

A number of ways have been suggested to curb the current rate of pollution. Such attempts at cleaning up the environment require plenty of time and resources to be pitched in. Industries have been given regulations for the disposal of hazardous waste, which aims at minimizing the area that becomes polluted. Organic methods of farming are being supported, which do not use chemical-laden pesticides and fertilizers. Use of plants that can remove the pollutants from the soil is being encouraged. However, the road ahead is quite long and the prevention of soil pollution will take many more years.

MUNICIPAL SOLID WASTE MANAGEMENT

Solid Waste Management may be defined as the discipline associated with the control of generation, collection, storage, transfer and transport, processing and disposal of solid wastes in a manner that is in accord with the best principles of public health, economics, engineering, conservation, aesthetics and other environmental considerations.

The most commonly recognized methods for the final disposal of solid wastes are:

Dumping on land

Dumping in water

Ploughing into the soil
Incineration

Municipal solid waste includes commercial and domestic wastes generated in municipal or notified areas in either solid or semi-solid form excluding industrial hazardous wastes but including treated bio-medical wastes. Municipal solid waste management (MSWM) is one of the major environmental problems of Indian cities. Improper management of municipal solid waste (MSW) causes hazards to inhabitants. Various studies reveal that about 90% of MSW is disposed of unscientifically in open dumps and landfills, creating problems to public health and the environment. In the present study, an attempt has been made to provide a comprehensive review of the characteristics, generation, collection and transportation, disposal and treatment technologies of MSW practiced in India.

CLASSIFICATION OF SOLID WASTES

(i) Domestic/Residential Waste:

This category of waste comprises the solid wastes that originate from single and multi-family household units. These wastes are generated as a consequence of household activities such as cooking, cleaning, repairs, hobbies, redecoration, empty containers, packaging, clothing, old books, writing/new paper, and old furnishings. Households also discard bulky wastes such as furniture and large appliances which cannot be repaired and used.

(ii) Municipal Waste:

Municipal waste includes wastes resulting from municipal activities and services such as street waste, dead animals, market waste and abandoned vehicles. However, the term is commonly applied in a wider sense to incorporate domestic wastes, institutional wastes and commercial wastes.

(iii) Commercial Waste:

Included in this category are solid wastes that originate in offices, wholesale and retail stores, restaurants, hotels, markets, warehouses and other commercial establishments. Some of these wastes are further classified as garbage and others as rubbish.

(iv) Institutional Waste:

Institutional wastes are those arising from institutions such as schools, universities, hospitals and research institutes. It includes wastes which are classified as garbage and rubbish as well as wastes which are considered to be hazardous to public health and to the environment.

(v) Garbage:

Garbage is the term applied to animal and vegetable wastes resulting from the handling, storage, sale, preparation, cooking and serving of food. Such wastes contain putrescible organic matter, which produces strong odours and therefore attracts rats, flies and other vermin. It requires immediate attention in its storage, handling and disposal.

(vi) Rubbish:

Rubbish is a general term applied to solid wastes originating in households, commercial establishments and institutions, excluding garbage and ashes.

(vii) Ashes:

Ashes are the residues from the burning of wood, coal, charcoal, coke and other combustible materials, for cooking and heating in houses, institutions and small industrial establishments. When produced in

large quantities at power generating plants and factories these wastes are classified as industrial wastes. Ashes consist of a fine powdery residue, cinders and clinker often mixed with small pieces of metal and glass.

(viii) Bulky Wastes:

In this category are bulky household wastes which cannot be accommodated in the normal storage containers of households. For this reason they require special collection. In developed countries bulky wastes are large household appliances such as cookers, refrigerators and washing machines as well as furniture, crates, vehicle parts, tyres, wood, trees and branches. Metallic bulky wastes are sold as scrap metal but some portion is disposed of at sanitary landfills.

(ix) Street Sweeping:

This term applies to wastes that are collected from streets, walkways, alleys, parks and vacant lots. In the more affluent countries manual street sweeping has virtually disappeared but it still commonly takes place in developing countries, where littering of public places is a far more widespread and acute problem. Mechanised street sweeping is the dominant practice in the developed countries. Street wastes include paper, cardboard, plastic, dirt, dust, leaves and other vegetable matter.

(x) Dead Animals:

This is a term applied to dead animals that die naturally or accidentally killed. This category does not include carcass and animal parts from slaughterhouses which are regarded as industrial wastes. Dead animals are divided into two groups, large and small. Among the large animals are horses, cows, goats, sheep, hogs and the like. Small animals include dogs, cats, rabbits and rats. The reason for this differentiation is that large animals require special equipment for lifting and handling during their removal. If not collected promptly, dead animals are a threat to public health because they attract flies and other vermin as they putrefy. Their presence in public places is particularly offensive and emits foul smell from the aesthetic point of view.

(xi) Construction and Demolition Wastes:

Construction and demolition wastes are the waste materials generated by the construction, refurbishment, repair and demolition of houses, commercial buildings and other structures. It mainly consists of earth, stones, concrete, bricks, lumber, roofing materials, plumbing materials, heating systems and electrical wires and parts of the general municipal waste stream, but when generated in large amounts at building and demolition sites, it is generally removed by contractors for filling low lying areas and by urban local bodies for disposal at landfills.

(xii) Industrial Wastes:

In the category are the discarded solid material of manufacturing processes and industrial operations. They cover a vast range of substances which are unique to each industry. For this reason they are considered separately from municipal wastes. It should be noted, however, that solid wastes from small industrial plants and ash from power plants are frequently disposed of at municipal landfills. For details please refer to Chapter 6 on “Industrial Wastes”.

(xiii) Hazardous Wastes:

Hazardous wastes may be defined as wastes of industrial, institutional or consumer origin which, because of their physical, chemical or biological characteristics are potentially dangerous to humans and the environment. In some cases although the active agents may be liquid or gaseous, they are classified as solid wastes because they are confined in solid containers. Typical examples are: solvents, paints and pesticides whose spent containers are frequently mixed with municipal wastes and become part of the urban waste stream. Certain hazardous wastes cause explosions in incinerators and fires at landfill sites. Others, such as pathological wastes from hospitals and radioactive wastes, require special handling at all time. Good management practice should ensure that hazardous wastes are stored, collected, transported and disposed off separately, preferably after suitable treatment to render them innocuous.

(xiv) Sewage Wastes:

The solid by-products of sewage treatment are classified as sewage wastes. They are mostly organic and derive from the treatment of organic sludge from both the raw and treated sewage. The inorganic fraction of raw sewage such as grit is separated at the preliminary stage of treatment, but because it entrains putrescible organic matter which may contain pathogens, must be buried/disposed of without delay. The bulk of treated, dewatered sludge is useful as a soil conditioner but invariably its use for this purpose is uneconomical. The solid sludge therefore enters the stream of municipal wastes unless special arrangements are made for its disposal.

CHARACTERISTICS OF MSW

Physical characteristics

Density

Density of waste, i.e., its mass per unit volume (kg/m^3), is a critical factor in the design of a solid waste management system, e.g., the design of sanitary landfills, storage, types of collection and transport vehicles, etc. To explain, an efficient operation of a landfill demands compaction of wastes to optimum density. Any normal compaction equipment can achieve reduction in volume of wastes by 75%, which increases an initial density of 100 kg/m^3 to 400 kg/m^3 . In other words, a waste collection vehicle can haul four times the weight of waste in its compacted state than when it is uncompacted. Significant changes in density occur spontaneously as the waste moves from source to disposal, due to scavenging, handling, wetting and drying by the weather and decomposition

Moisture content

Moisture content is defined as the ratio of the weight of water (wet weight - dry weight) to the total wet weight of the waste. Moisture increases the weight of solid wastes, and thereby, the cost of collection and transport. In addition, moisture content is a critical determinant in the economic feasibility of waste treatment by incineration, because wet waste consumes energy for evaporation

of water and in raising the temperature of water vapour. In the main, wastes should be insulated from rainfall or other extraneous water. We can calculate the moisture percentage, using the formula given below

A typical range of moisture content is 20 to 40%, representing the extremes of wastes in an arid climate and in the wet season of a region of high precipitation. However, values greater than 40% are not uncommon. Climatic conditions apart, moisture content is generally higher in low income countries because of the higher proportion of food and yard waste.

Size of Waste constituents

The size distribution of waste constituents in the waste stream is important because of its significance in the design of mechanical separators and shredder and waste treatment process. This varies widely and while designing a system, proper analysis of the waste characteristics should be carried out.

Calorific Value

Calorific value is the amount of heat generated from combustion of a unit weight of a substance, expressed as kcal/kg. The calorific value is determined experimentally using Bomb calorimeter in which the heat generated at a constant temperature of 25°C from the combustion of a dry sample is measured.

The physical properties that are essential to analyse of wastes disposed at landfills are:

Field capacity

The field capacity of municipal solid waste is the total amount of moisture which can be retained in a waste sample subject to gravitational pull. It is a critical measure because water in excess of field capacity will form leachate, and leachate can be a major problem in landfills. Field capacity varies with the degree of applied pressure and the state of decomposition of the wastes.

Permeability of compacted wastes

The hydraulic conductivity of compacted wastes is an important physical property because it governs the movement of liquids and gases in a landfill. Permeability depends on the other properties of the solid material include pore size distribution, surface area and porosity. Porosity represents the amount of voids per unit total volume of material. The porosity of municipal solid waste varies typically from 0.40 to 0.67 depending on the compaction and composition of the waste.

Compressibility

It is the degree of physical changes of the suspended solids or filter cake when subjected to pressure.

Chemical characteristics

Knowledge of the classification of chemical compounds and their characteristics is essential for the proper understanding of the behaviour of waste, as it moves through the waste management

system. The products of decomposition and heating values are two examples of chemical characteristics. If solid wastes are to be used as fuel, or are used for any other purpose, we must know their chemical characteristics, including the following

Chemical: Chemical characteristics include pH, Nitrogen, Phosphorus and Potassium, total Carbon, C/N ratio, calorific value.

Bio-Chemical: Bio-Chemical characteristics include carbohydrates, proteins, natural fibre, and biodegradable factor.

Toxic: Toxicity characteristics include heavy metals, pesticides, insecticides, Toxicity test for Leachates (TCLP), etc.

Lipids

This class of compounds includes fats, oils and grease. Lipids have high calorific values, about 38000 kcal/kg, which makes waste with a high lipid content suitable for energy recovery processes. Since lipids in the solid state become liquid at temperatures slightly above ambient, they add to the liquid content during waste decomposition. They are biodegradable but because they have a low solubility in waste, the rate of biodegradation is relatively slow.

Carbohydrates

Carbohydrates are found primarily in food and yard waste. They include sugars and polymers of sugars such as starch and cellulose and have the general formula $(CH_2O)_x$. Carbohydrates are readily biodegraded to products such as carbon dioxide, water and methane. Decomposing carbohydrates are particularly attractive for flies and rats and for this reason should not be left exposed for periods longer than is necessary.

Proteins

Proteins are compounds containing carbon, hydrogen, oxygen and nitrogen and consist of an organic acid with a substituted amine group (NH_2). They are found mainly in food and garden wastes and comprise 5-10% of the dry solids in solid waste. Proteins decompose to form amino acids but partial decomposition can result in the production of amines, which have intensely unpleasant odours.

Natural fibres

This class includes the natural compounds, cellulose and lignin, both of which are resistant to biodegradation. They are found in paper and paper products and in food and yard waste. Cellulose is a larger polymer of glucose while lignin is composed of a group of monomers of which benzene is the primary member. Paper, cotton and wood products are 100%, 95% and 40% cellulose respectively. Since they are highly combustible, solid waste having a high proportion of paper and wood products, are suitable for incineration. The calorific values of oven-dried paper products are in the range 12000 – 18000 kcal/kg and of wood about 20000 kcal/kg, which compare with 44200 kcal/kg for fuel oil.

Synthetic organic material (Plastics)

They are highly resistant to biodegradation and, therefore, are objectionable and of special concern in solid waste management. Hence the increasing attention being paid to the recycling of plastics to reduce the proportion of this waste component at disposal sites. Plastics have a high heating value, about 32,000 kJ/kg, which make them very suitable for incineration. But, one should note that polyvinyl chloride (PVC), when burnt, produces dioxin and acid gas. The latter increases corrosion in the combustion system and is responsible for acid rain.

Non-combustibles:

This class includes glass, ceramics, metals, dust and ashes, and accounts for 12 – 25% of dry solids.

Heating value

An evaluation of the potential of waste material for use as fuel for incineration requires a determination of its heating value, expressed as kilojoules per kilogram (kJ/kg). The heating value is determined experimentally using the *Bomb calorimeter test*, in which the heat generated, at a constant temperature of 25°C from the combustion of a dry sample is measured. Since the test temperature is below the boiling point of water (100°C), the combustion water remains in the liquid state. However, during combustion, the temperature of the combustion gases reaches above 100°C, and the resultant water is in the vapour form. While evaluating incineration as a means of disposal or energy recovery, one has to consider the heating values of respective constituents.

Ultimate analysis

This refers to an analysis of waste to determine the proportion of carbon, hydrogen, oxygen, nitrogen and sulphur, and it is done to perform mass balance calculation for a chemical or thermal process. Besides, it is necessary to determine ash fraction because of its potentially harmful environmental effects, brought about by the presence of toxic metals such as cadmium, chromium, mercury, nickel, lead, tin and zinc. One should note that other metals (e.g., iron, magnesium, etc.) may also be present but they are non-toxic. C, H, N, S, O is analysed.

Proximate analysis

This is important in evaluating the combustion properties of wastes or a waste or refuse derived fuel. The fractions of interest are:

- moisture content, which adds weight to the waste without increasing its heating value, and the evaporation of water reduces the heat released from the fuel;
- ash, which adds weight without generating any heat during combustion;
- volatile matter, i.e., that portion of the waste that is converted to gases before and during combustion;
- fixed carbon, which represents the carbon remaining on the surface grates as charcoal. A waste or fuel with a high proportion of fixed carbon requires a longer retention time on the

furnace grates to achieve complete combustion than a waste or fuel with a low proportion of fixed carbon.

Moisture content, Volatile matters, Fixed carbon and Ash is estimated in proximate analysis.

Collection and Transport

<https://www.open.edu/openlearncreate/mod/oucontent/view.php?id=80575&printable=1>

Littering of municipal solid waste shall be prohibited in cities, towns and in urban areas notified by the State Governments. To prohibit littering and facilitate compliance, the following steps shall be taken by the municipal authority, namely

- **Organizing house-to-house** collection of municipal solid wastes through any of the methods, like community bin collection (central bin), house-to-house collection, collection on regular pre-informed timings and scheduling by using musical bell of the vehicle.
- Devising collection of waste from slums and squatter areas or localities including hotels, restaurants, office complexes and commercial areas.
- Wastes from slaughter houses, meat and fish markets, fruits and vegetable markets, which are biodegradable in nature, shall be managed to make use of such wastes.
- Bio-medical wastes and industrial wastes shall not be mixed with municipal solid wastes and such wastes shall follow the rules separately specified for the purpose.
- Collected waste from residential and other areas shall be transferred to community bin by hand-driven carts or other small vehicles.
- Construction or demolition wastes or debris shall be separately collected and disposed off following proper norms. Similarly, wastes generated at dairies shall be regulated in accordance with the State laws.
- Waste (garbage, dry leaves) shall not be burnt.
- Stray animals shall not be allowed to move around waste storage facilities or at any other place in the city or town.

Storage of municipal solid wastes

Processing of municipal solid wastes

Municipal authorities shall adopt suitable technology or combination of such technologies to make use of wastes so as to minimize burden on landfill. Following criteria shall be adopted, namely

- The biodegradable wastes shall be processed by composting, vermicomposting, anaerobic digestion or any other appropriate biological processing for stabilization of wastes.
- Mixed waste containing recoverable resources shall follow the route of recycling.

- Incineration with or without energy recovery can also be used for processing wastes in specific cases.

Disposal of municipal solid wastes

Land filling shall be restricted to non-biodegradable, inert waste and other waste that are not suitable either for recycling or for biological processing. Land filling shall also be carried out for residues of waste processing facilities as well as pre-processing rejects from waste processing facilities. Land filling of mixed waste shall be avoided unless the same is found unsuitable for waste processing. Under unavoidable circumstances or till installation of alternate facilities, land filling shall be done following proper norms.

SANITARY LANDFILLS

In 1935, a new system of waste disposal, called sanitary landfills, was created in Fresno, California. Currently, over 55% of all municipal solid waste that is created in the United States is disposed of in sanitary landfills. **Sanitary landfills** are a method of waste disposal where the waste is buried either underground or in large piles. This method of waste disposal is controlled and monitored very closely. The main purpose of sanitary landfill is to ensure waste is safe by reducing the harm from accumulated waste and allowing safe decomposition. The alternative layering of garbage and soil aids in hastening decomposition. Because methane, a toxic gas, is produced from decomposition, it is collected in the landfill to generate electricity instead of being released into the atmosphere. The sanitary landfill also uses a clay liner to isolate the trash from the environment. Besides, it involves well-designed engineering methods to protect the environment from contamination by solid or liquid wastes. When the land used as the sanitary landfill is full, impervious clay is used to seal it and if deemed safe, that area can be used for other purposes.

How Sanitary Landfill Works?

Sanitary landfills operate by layering waste in a large hole. There are different levels of layering of waste to facilitate decomposition of the materials as well as trap toxic gases released from the process. The layers are made in such a way that the bottom part has the smallest volume with the top part taking the bigger volume to avoid collapsing of the land.

The first layer is the liner system. This is the lowest layer and the first one used in laying a foundation for the sanitary landfill. The process entails the application of compact and well dense clay to prevent the seeping in or out of liquids. It is for this reason that this clay is completely impervious.

This modernized type of sanitary landfill is also fitted with high density plastic which is applied on top of the clay for reinforcement. This engineering design is to make certain there is no room for the penetration of liquids thus preventing contamination of the underground water.

The second layer is the drainage system. The drainage system takes care of the liquid produced from the decomposition of some waste materials. Due to the toxicity of this liquid, it should not seep past the liner layer. The drainage layer aids in draining away this liquid to avoid it getting close to the liner system. Moreover, rainfall and snow may also seep from the top surface to the landfill and also needs to be drained away.

These liquids may contain contaminants from the waste materials that can corrode the liner system and contaminate the soil and water below the landfill. In reducing this risk, the sanitary landfill is fitted with perforated pipes on top of the liner layer to collect all the liquids that find their way to the bottom of the landfill through leaching thus the name leachate. The plumbing system then directs the leachate to treatment plants to undergo treatment for reuse.

The third layer is the gas collection system. In the same way that liquids are produced, gases are also released through natural processes. Methane is the most common of these gases. Methane is a gaseous hydrocarbon which is very volatile as well as toxic and thus, when released into the atmosphere, it contributes to global warming and can harm the overall environmental health.

There are, therefore, extraction pipes in this layer that trap methane gas and transport it to treatment plants to treat the gas and thereafter, use it to produce electricity and to power various processes.

The fourth layer contains the trash itself. This is the largest layer and is also the topmost. Periodically, trash from various sources is brought in by the various garbage collection companies and dumped in this layer. To avoid taking too much space, the garbage disposed is compacted on a daily basis.

Once this has been done, a layer of compacted soil is applied on the surface of the sanitary landfill. Soil does a good job of containing bad smells and the growth of harmful microorganisms such as pests and flies. Also, the compact soil keeps away windblown debris.

Landfills have been known to be as deep as hundreds of feet and can take years before they are fully filled after compaction. In case a sanitary landfill gets filled, landfill capping is done. This is the process of capping or topping the landfill with synthetic plastic or clay, just as the bottom of the landfill. The aim of this is to prevent methane gas from escaping to the atmosphere as well as bad odors.

Further, this layer is reinforced with a layer of top soil, about 2-3 feet and vegetation is planted. When all this is completed, the land is reclaimed and can be used for other purposes. In spite of all these processes, due to the toxicity of the waste buried deep down, there is still a chance of contamination, either of the water ways or the soil which affects plants, some of which are edible to both man and animals.

To avoid this, sanitary landfills that have been closed and repurposed for gardening or any other form of crop planting where they are monitored for decades and the soil regularly tested for irregularities. If the plants are dying then there might still be methane gas being released from the land. Once it is proved that the land is safe, the area is open for other various uses.

Heavy duty projects such as construction of housing units are not at all allowed since the ground will have to be scrapped for the establishment of a suitable foundation. Simple land intensive projects such as open spaces, nature parks and golf courses are encouraged on such lands.

TECHNOLOGIES FOR PROCESSING OF MSW

1.INCINERATION

Incineration is the treatment of waste material by combustion of organic substances present in the waste materials. It converts the waste material into heat, flue gas and ash which are released into the atmosphere without any further treatment for usage. Heat which is in the major percentage can be used to generate electric power. Flue gases contain traces of nitrogen, carbon dioxide and sulfur dioxide, each of which has better utilization when used optimally. Nitrogen produced can be used as fertilizers to increase the productivity of crops, carbon dioxide can be used as fire extinguishers and sulfur once extracted from sulfur dioxide can be used in dental treatment. Ash is obtained in the form of solid lumps which can be used for construction purposes. The benefit of incineration is that it reduces the solid mass of organic wastes by 80-85% and volume by 95-96% which is commendable. A bottom ash is the most significant by-product from MSWI, which accounts for about 80%-87% (in weight) of the solid residues. The amount of bottom ash is 20%-30% of the initial weight of wastes incinerated. Due to its origin, MSWI bottom ash is mainly composed of high-temperature solids, which is rapidly cooled down when the material is quenched after exiting the combustion chamber. It is mainly composed of Si, Fe, Ca, Al, Na and K, in the form of oxides, and thus, presents a similar composition to that of geological materials. From the constitutional phases of MSWI bottom ash, it is therefore similar to the geological materials. A typical scheme for an incineration plant operating on real waste and with energy recovery is given below-

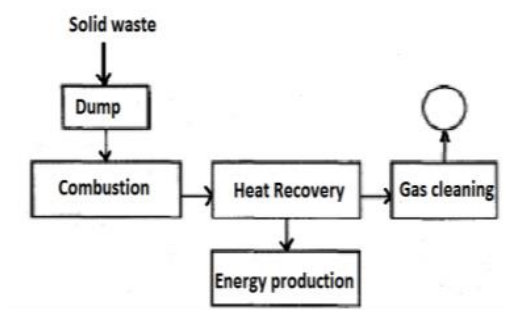
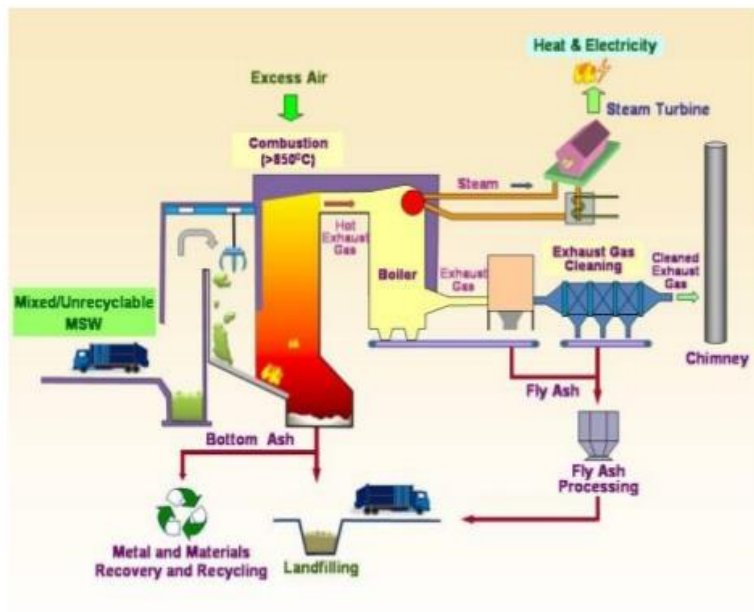


Fig-1. Flow chart of incineration process

The collected material is located in a land site and from here taken to be sent to the incineration process, normally performed in grid or rotating kilns. The gas coming from the combustion process at a temperature around 900o C to 1000o C is sent to a heat exchanger for steam production to be then used in a thermal cycle for energy production. Out of the exchanger, at a temperature around 200o to 250oC,

the gas is then sent to the gas cleaning line to abate the polluting substances (dust, acid gas, etc.), and is discharged from the chimney into the atmosphere. As already noted, energy recovery is performed in a thermal cycle, usually with a turbine and condenser to maximize the production of electric energy.

2. INCINERATION PROCESS:



Combustion: Waste is continuously fed into the furnace by an overhead crane. The waste is combusted in the specially designed furnace at high temperature of $> 850^{\circ}\text{C}$ for more than 2 second with sufficient supply of air so as to ensure complete burning of the waste and to prevent the formation of dioxins and carbon monoxide.

Boiler/ steam turbine: The heat from the combustion is used to generate steam in the boiler. The steam then drives the turbine which is coupled to the electricity generator. The excess heat generated can also be used for other purposes, e.g. heat for swimming pool.

Exhaust gas cleaning: The exhaust gas from the boiler is typically cleaned by the following advanced pollution control systems to ensure compliance with the stringent environmental standards.

Dry or Wet scrubbers: To spray lime powder or fine atomized slurry into the hot exhaust gas to neutralize and remove the polluted acidic gases (sulphur oxides, hydrogen chloride).

Activated Carbon Injection: To adsorb and remove any heavy metal and organic pollutants (e.g. dioxins) in the exhaust gas.

Bag house filter: To filter and remove dust and fine particulates.

Selective Non-Catalytic Reduction: To remove a nitrogen oxide (which is a cause of urban smog) by reacting them with ammonia or urea

3. TYPES OF INCINERATORS

There are three main types of combustion technologies in commercial practice:

1. Rotary Kiln,
2. Moving Grate

3. Fluidized Bed

2.COMPOSTING

Another method of treating municipal solid waste is composting, a biological process in which the organic portion of refuse is allowed to decompose under carefully controlled conditions. Microbes metabolize the organic waste material and reduce its volume by as much as 50 percent. The stabilized product is called compost or humus. It resembles potting soil in texture and odour and may be used as a soil conditioner or mulch.

Composting offers a method of processing and recycling both garbage and sewage sludge in one operation. As more stringent environmental rules and siting constraints limit the use of solid-waste incineration and landfill options, the application of composting is likely to increase. The steps involved in the process include sorting and separating, size reduction, and digestion of the refuse.

Sorting and shredding

The decomposable materials in refuse are isolated from glass, metal, and other inorganic items through sorting and separating operations. These are carried out mechanically, using differences in such physical characteristics of the refuse as size, density, and magnetic properties. Shredding or pulverizing reduces the size of the waste articles, resulting in a uniform mass of material. It is accomplished with hammer mills and rotary shredders.

Digesting and processing

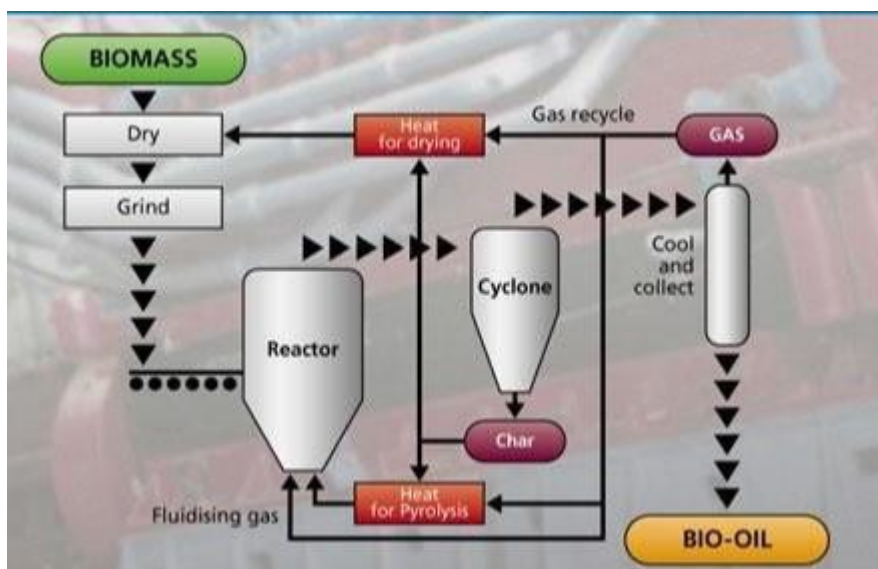
Pulverized waste is ready for composting either by the open windrow method or in an enclosed mechanical facility. Windrows are long, low mounds of refuse. They are turned or mixed every few days to provide air for the microbes digesting the organics. Depending on moisture conditions, it may take five to eight weeks for complete digestion of the waste. Because of the metabolic action of aerobic bacteria, temperatures in an active compost pile reach about 65 °C (150 °F), killing pathogenic organisms that may be in the waste material.

Open windrow composting requires relatively large land areas. Enclosed mechanical composting facilities can reduce land requirements by about 85 percent. Mechanical composting systems employ one or more closed tanks or digesters equipped with rotating vanes that mix and aerate the shredded waste. Complete digestion of the waste takes about one week. Digested compost must be processed before it can be used as a mulch or soil conditioner. Processing includes drying, screening, and granulating or pelletizing. These steps improve the market value of the compost, which is the most serious constraint to the success of composting as a waste management option. Agricultural demand for digested compost is

usually low because of the high cost of transporting it and because of competition with inorganic chemical fertilizers.

3.PYROLYSIS

Pyrolysis is rapidly developing biomass thermal conversion technology and has been garnering much attention worldwide due to its high efficiency and good eco-friendly performance characteristics. Pyrolysis technology provides an opportunity for the conversion of municipal solid wastes, agricultural residues, scrap tires, non-recyclable plastics etc into clean energy. It offers an attractive way of converting urban wastes into products which can be effectively used for the production of heat, electricity and chemicals.



Pyrolysis of Municipal Wastes

Pyrolysis process consists of both simultaneous and successive reactions when carbon-rich organic material is heated in a non-reactive atmosphere. Simply speaking, pyrolysis is the thermal degradation of organic materials in the absence of oxygen. Thermal decomposition of organic components in the waste stream starts at 350°C–550°C and goes up to 700°C–800°C in the absence of air/oxygen.

Pyrolysis of municipal wastes begins with mechanical preparation and separation of glass, metals and inert materials prior to processing the remaining waste in a pyrolysis reactor. The commonly used pyrolysis reactors are rotary kilns, rotary hearth furnaces, and fluidized bed furnaces. The process requires an external heat source to maintain the high temperature required.

Pyrolysis can be performed at relatively small-scale which may help in reducing transport and handling costs. In pyrolysis of MSW, heat transfer is a critical area as the process is endothermic and sufficient heat transfer surface has to be provided to meet process heat requirements.

The main products obtained from pyrolysis of municipal wastes are a high calorific value gas (synthesis gas or syngas), a biofuel (bio oil or pyrolysis oil) and a solid residue (char). Depending on the final temperature, MSW pyrolysis will yield mainly solid residues at low temperatures, less than 4500C, when the heating rate is quite slow, and mainly gases at high temperatures, greater than 8000C, with rapid heating rates. At an intermediate temperature and under relatively high heating rates, the main product is a liquid fuel popularly known as bio oil.

What are the Advantages of Pyrolysis?

The key benefits of pyrolysis include the following:

- It is a simple, inexpensive technology for processing a wide variety of feedstocks.
- It reduces waste going to landfill and greenhouse gas emissions.
- It reduces the risk of water pollution.
- It has the potential to reduce the country's dependence on imported energy resources by generating energy from domestic resources.
- Waste management with the help of modern pyrolysis technology is inexpensive than disposal to landfills.
- The construction of a pyrolysis power plant is a relatively rapid process.
- It creates several new jobs for low-income people based on the quantities of waste generated in the region, which in turn provides public health benefits through waste clean up.